



TOWARDS EFFECTIVE DESIGN AND ADAPTATION OF CSP USING MODELLING AND SIMULATION BASED DIGITAL TWIN APPROACH

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ABSTRACT

Modern communication service providers (CSPs) continuously innovate their product offerings, and adapt engagement services for sales and care processes to increase customer base, maximize customer lifecycle value (CLV) and reduce customer churn in order to stay ahead of the competition and remain profitable. These telecom companies chiefly need an ability to introduce right product offerings at the right time, identify what to cross-sell and upsell to whom, and make timely course correction on engagement services. The current practice of meeting these needs in experience-centric and intuition-driven manner is turning out to be ineffective in the increasingly dynamic business environment. This paper proposes a multi-modelling based simulatable digital twin that enables in-silico quantitative exploration of design space to help human experts arrive at the right product offerings and engagement services. This paper describes our approach and illustrates its utility using a real-life case study.

Keywords: Digital Twin; Design; Adaptation; Adaptable Enterprise;

1 INTRODUCTION

Communication service providers (CSPs) operate in a dynamic and uncertain business environment where they inevitably experience various kinds of disruptions, such as introduction of 5G technology and new mobile devices, paradigm shift in user expectations, and competition in the space of product offerings and engagement services (Maljevic et al. 2019). A successful CSP needs to continuously offer innovative products and adapt their supporting services in response to the various change drivers so as to minimize churn while attracting new customers. Failure to do so negative impacts key business performance indicators, which may turn out to be irreversible considering the highly competitive situation.

To come up with new products and different engagement services that include sales and care services, CSPs rely on the insights and intuitions borne out of human expertise, past-experience and controlled experimentation. For example, they often rely on ‘*spray and pray*’ on potential customer target groups for designing and piloting a new product in the market, and ‘*trial and error*’ iterations for adaptation and optimization of sale and customer care services. This iterative process with one or multiple ineffective exploration choice leads to possibly irreversible customer churn and subsequent revenue loss. Moreover, high attrition rate depletes corporate knowledge bank thereby further elongating the payback cycles. Thus, inefficiencies in designing new products / refining existing products, and associated customer care services significantly impact the top-line as well as the bottom-line of CSPs – undesirable situation especially in the highly competitive and rapidly transforming telecom market.

The essential business requirement is to come up with answers for questions such as: Given the set of products and engagement services, what’s the best *Average Revenue Per User* (ARPU), *Customer Lifecycle Value* (CLV) and *Net Promoter Score* (NPS) possible for the given customer base? What changes need to be introduced in products and/or engagement services in order to attain the desired key performance indicator (KPI) values (*i.e.*, ARPU, CLV, NPS, *etc.*)? Which of the existing products should be retired? Which customer segments to prioritize over other low yielding segments to improve margins? Which product / customer experience to upsell to whom and when to increase the price point? And so on. In a sense, the problem appears to be a complex multi-variate optimization problem with large number of variables pertaining to products, customers, and engagement services.

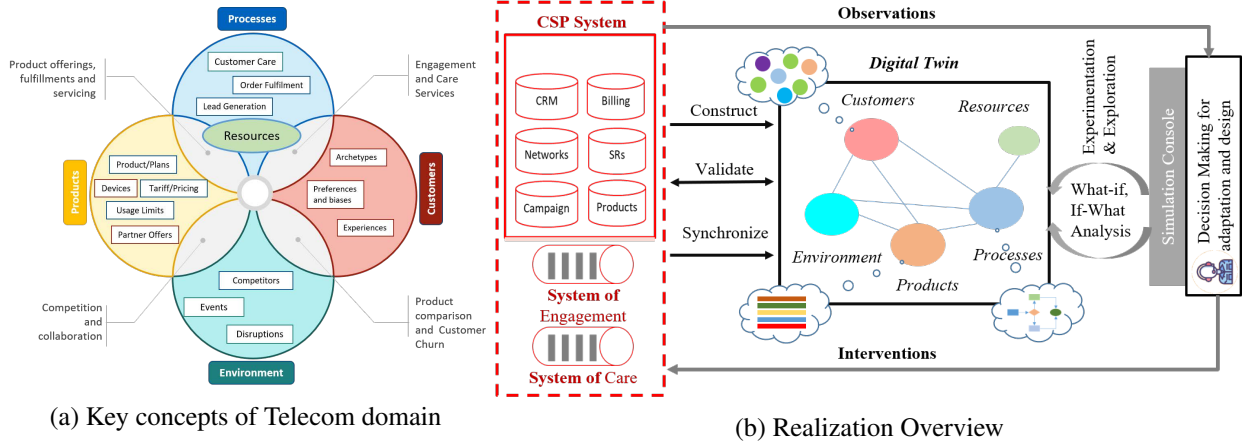


Figure 1: Key concepts and a realization

However, purely analytical techniques are not a pragmatic consideration here as the overall system behavior is not a-priori knowable. Conceptually, a CSP is characterized as a *system of systems* (Boardman and Sauser 2006) with multiple non-linear dependencies, uncertainties and emergentism as opposed to a monolithic unit with static and known behaviour.

Recent trend is to use machine learning techniques on past data to arrive at a model of the reality. The disadvantage of relying solely on machine learnt model is that it's a subset of the comprehensive model as the data represents only the system behaviours that have manifested but not necessarily all possible behaviours the system is capable of. Moreover, the relevancy of the past data reduces over time for dynamic operating environment. Naturally, sole dependency on the analyses derived from such a subset model results into imprecise and probably even incorrect interpretation.

We present a multi-modelling based simulation-driven digital twin approach to help human experts understand (*i.e.*, what is the current state and why) and explore adaptation (*i.e.*, what is the right intervention for a specific perturbation) and design (*i.e.*, introduce new business offerings to operate in an environment effectively) alternatives. For example, a CSP can now iteratively experiment the uptake by their customer base and ARPU generated against existing product catalog with adapted service offerings, new product as well as engagement strategies using a virtual environment. Thus, this approach enables cost-effective in-silico experimentation to answer the questions listed earlier in a quantitative justification-backed manner.

In our approach, we adopt the concept of digital twin (Boschert and Rosen 2016) as an overarching approach, and build further upon integrating the concepts of actor model of computation (Agha 1986) and multi-criteria decision making (MCDM) (Triantaphyllou 2000) to specify digital twin of CSP. We use a proven validation approach, termed as operation validity (Sargent 2005), to establish the faithfulness of the constructed digital twin, and rely upon an actor-based bottom-up simulation engine, ESL simulator (Clark et al. 2017), to understand a CSP and explore adaptation and design of product, customer and process related alternatives to improve relevant KPIs of a CSP.

This paper presents our approach, contextualizes it for a class of business problems from telecom industry and demonstrates how effectively our approach helps domain experts to explore and decide right interventions at right time with improved certainty. Paper further makes a claim that an effective use of simulation-driven digital twin in designing and adapting products and processes in telecom domain can be as beneficial as other established domains such as mission-critical systems (Glaessgen and Stargel 2012) and engineering field (Schleich et al. 2017). The rest of the paper is organised as follows: section 2 discusses the problem domain and represents the domain understanding in terms of a conceptual (meta-) model. Section 3 discusses the state-of-the-practice and state-of-the-art of modelling and simulation techniques that can be considered for developing a digital twin of a complex enterprise such as a telecom company. It also demonstrates the limitations of the state-of-the-art modelling and analysis techniques. Section 4 presents our digital twin approach and highlights a method to construct, validate and simulate what-if scenarios. In section 5, we illustrate efficacy of the proposed approach through a real life business use case. We conclude discussing our early experience of using digital twin in telecom business and outlining future work in section 6.

2 PROBLEM DOMAIN

A typical communication service provider (CSP) can be described using three internal elements namely: *product*, *customer* and *process*, and an external element, which we term as *environment*, as shown in Figure 1a. A *product* of a CSP is a bundled offering of a end user computing device, telecommunication services and partner services such as OTT platform streaming. The *customers* of a CSP are the individuals or business entities, with varying usage and behavioural characteristics, who subscribe one or more products and use them for a specific period. The processes realise two types engagement services of a CSP: *sales service* and *care service*. The sales services help a CSP to engage with the existing and/or potential customers through appropriate channels, such as store, web and phone. The examples of sales services are: product sales, promotions and offers. The care services help to assist customers throughout their life cycle of the products and resolve the concerns related to order fulfilment and product usage, such as billing and network related issues. The external element, *i.e.*, *environment*, of a CSP represents all external factors that include its *competitions* and *external events* that impact CSP's operation, such as technological advancement and adoption (*e.g.*, adoption of 5G technology), weather, festivals, sports, cultural events and other relevant social gatherings.

For telecommunication services, a customer engages with a CSP for a product or a set of products by evaluating the product offerings from the CSP and its competitors. The sales services typically nudge the evaluation process of individuals. Once the sale is completed (*i.e.*, product is ordered and activated), a customer experiences product life cycle journey with the CSP, where care services play a major role to resolve customer issues and concerns. For a CSP, the product offerings and sales services together help to increase customer base. The care services decide the customer churn tendencies. A suitable combination of product offering, sales services and care services for an environment decide most of the key performance indicators (KPIs) of a CSP, such as average revenue per user (ARPU), customer lifecycle value (CLV), steady customer base, net promoter score (NPS) and loyalty. For a CSP, inability to introduce suitable offerings on time, redefine existing offerings in response to environment change, and make timely and proactive course correction of sales and care services significantly impact the top-line and bottom-line.

Therefore, the crucial factors affecting CSP performance are: (i) offer attractive products such that more number of customers find them useful and start using them, (ii) design/adapt sales services to reach out more customers and nudge them suitably, and (iii) provide adequate care service so that existing customers stay happy and connected with the CSP. However, coming up with new innovative offerings for a wide spectrum of customers, adapting existing offerings to meet the emerging needs of customers in a dynamic environment, and attempting to provide the necessary care while facing equally innovative and adaptive competition are significant challenges for today's CSPs.

From system theoretic perspective, we relate a CSP as complex *system of systems* (Boardman and Sauser 2006) as it comprises a wide range of products with varying characteristics and life spans, large number of customers with autonomous and probabilistic behaviours, and a set of adaptable processes. The processes have their own goals and the goals are realised by dynamically orchestrating a set of activities, which are performed by human and/or machines (owned by different units or departments). Overall a CSP system (of system) interacts with an uncertain environment, where the competitors of the CSP continuously adapt in terms of product, sales and care services, and multiple stochastic events take place over the time. While operating in a dynamic and uncertain environment, a CSP system exhibits multiple (non-linear) feedback loops (shown in interaction area of Figure 1a) between: (a) customers and environment, (b) product and environment, and (c) product offerings, customers and processes. Here, analysing overall system behaviour and exploring effective adaptation and design alternatives become difficult as constituent (sub-) systems exhibit significant uncertainty and non-linear feedback loops, and overall behaviour emerges from interaction of multiple dynamically connected (sub-) systems. Thus, the need is for continuous adaption in the face of partial information and uncertainty.

3 STATE OF THE ART OF MODELLING AND ANALYSIS

The analysis to understand a complex system is typically approached using two broad categories: qualitative approach and quantitative approach. The qualitative approach is concerned with the subjective assessment of the underlying system through a range of management techniques, such as interviews, discussions, and field studies. The state of the practice of analysing CSPs, *i.e.*, product and service design and adaptations, are chiefly based on such subjective and qualitative analyses (Oughton 2018). Subjective analyses those

are purely based on human expertise and intuitions often result into ineffective solution as Telecom domain is characterized by significant non-linearities, inherent uncertainties at multiple localized contexts, and the overall behaviour often emerges over time.

The quantitative approach, in contrast, involves precise analysis and interpretation of system data, structure and behaviour. The quantitative approach can be further classified into three categories (Kothari 2004): (i) inferential approach, (ii) experimental approach and (iii) modelling and simulation approach. The inferential approach (Michalski 1993) relies on the existing system data (*i.e.*, historical traces) to infer the characteristics of a system using established AI techniques. The state-of-the-art digital twin approaches pertaining to Telecom industry fundamentally rely on such AI based inferencing approach. An inferential approach fails to demonstrate desired efficacy for a system that operates in a dynamic environment. For example, prediction of an effective future products with 5G technology/devices by analyzing historical data of a CSP that has not observed the behavior of a customer base without 5G technology might not be an ideal situation.

The experimental approach comprehends a system by manipulating the system variables and observing their effects in a controlled environment. The experimental approaches, such as trial-and-error of care service on a selected customer group, may not be an effective option for telecom industry. For example, conducting experiments on effective care service with several customer touch points to a wide range of customers is not an economically viable option.

The modelling and simulation approaches rely on the philosophy of *science of the artificial* (Simon 1996) where the intent is: (i) to imitate the real system using its purposive model, (ii) to explore a range of scenarios by simulating the possible (forward looking) changes incorporated into the model, and (iii) to develop a precise understanding of the system by interpreting the simulation results. Based on the modelling paradigms, it can be further classified into two broad approaches: *top-down* approach and *bottom-up* approach (Thomas and McGarry 1994). A top-down approach models a system as a whole and adopts reductionist view to decompose it into smaller parts to understand the parts in isolation. This approach uses a range of models to represent and analyse systems, such as mathematical model and enterprise model (EM). The mathematical models *e.g.*, linear programming (Candes and Tao 2005) represent the system using mathematical formulae and use rigorous mathematical and statistical problem-solving techniques for system analysis. Over the years, Simulink and MATLAB (Klee and Allen 2016) are extensively used to model, analyse and simulate complex mission-critical systems. Such an approach is a useful consideration when relevant aspects of a system can be suitably specified using algebraic equations, which is a difficult proposition for telecom domain due to domain specific inherent complexities that include high dynamism, uncertainty and emergent behaviour. The enterprise models (EMs), such as ArchiMate (Iacob et al. 2012), are typically less rigorous than mathematical models, however, they are capable of representing the key aspects of the enterprises. The key limitation of such top-down modelling and simulation approaches is that they are not suitable for specifying adaptability and inherent uncertainty. Moreover, they are not cognisant of emergentism.

Agent and actor-based technologies, such as Erlang (Armstrong 1996), support a bottom-up approach that starts from the parts or micro-behaviours of the system and derives a holistic view of the system through composition. Conceptually, they fare better to model and analyse systems that are characterized by autonomy, reactivity, adaptability and emergentism. However, support for uncertainty and scalability (say millions of actors in a system) are some of the key limitations of the state-of-the art actor/agent technologies.

4 OUR APPROACH

We propose a pragmatic actor-based simulatable digital twin to support quantitative analysis for evaluating adaptation and design alternatives of a CSP (or its part) to improve its KPIs such as ARPU, CLV, customer base and NPS. A schematic overview of our approach is shown in Figure 1b. We first construct a high-fidelity simulatable digital twin, validate it, set it up with real data, and simulate various (hypothetical) adaptation and design alternatives to understand their impacts on KPIs. Our approach rests on four pillars: (i) *digital twin* (Boschert and Rosen 2016) to enable in-silico quantitative analysis, (ii) actor-based modelling paradigm (Agha 1986) to faithfully represent individual characteristics of CSP as a system of systems as close to the reality as possible, (iii) bottom-up simulation (Macal and North 2010, Barat et al. 2019) to support emergence of the overall behaviour of CSP, and (iv) two steps modelling and analysis method recommended by Robert Sargent in (Sargent 2005) to construct, validate and analyse a simulation model with appropriate methodological rigour.

4.1 Rationales of Design Choices

The digital twin concept, *i.e.*, a comprehensive and machine-interpretable replica of a real system (Boschert and Rosen 2016), is used extensively in mission-critical systems (Glaessgen and Stargel 2012) and engineering field (Schleich et al. 2017) for a wide range of interrogative and predictive analyses leading to decision making (Grieves and Vickers 2017). Primarily, they develop high-fidelity physics and mathematical models to represent system behaviours, and consider analytical simulation as an aid to evaluate the efficacy of a set of candidate changes for achieving system goals. Over the time, it is observed that such an approach is pragmatic and cost effective as it helps to understand the consequences of prospective adaptation and design choices prior to their implementations. Such an in-silico analysis reduces or eliminates real-life experiments, which are often expensive and infeasible. Despite the established economic benefit in engineering domain, digital twin based analysis has not received serious consideration in business systems. This is probably due to salient characteristics such as *system of systems* nature, autonomy of individual systems, partial information, uncertainty, and dynamism.

We consider *actor model of computation* (Agha 1986) as the primitive modelling abstraction to specify digital twin. We represent the CSP system of systems (*i.e.*, customers, products and processes) and its environment (*i.e.*, competitors and the events of interest) as a set of *actors* exchanging messages amongst them. Our *actor* abstraction is characterized by a set of *actor* characteristics that include *modularity*, *composability*, *reactiveness*, *autonomy*, and *adaptiveness*. This canonical actor abstraction is extended to support probabilistic behavior and multi-criteria decision making (MCDM) (Triantaphyllou 2000) capability. Probabilistic behaviour helps capture inherent uncertainty of the system, for example, on receiving an alert informing overuse of data the customer may resort to a set of actions exhibiting certain probability distribution to reduce data consumption. We use MCDM to represent and evaluate the point decision-making of an individual actor. For example, determining the propensity of buying a specific product (with product characteristics) from a list of alternatives by an individual customer (with specific affinity about product characteristics) at a specific time (*e.g.*, at the end of an existing subscription or introduction of a new product) can be effectively defined and evaluated using MCDM techniques, such as AHP (Triantaphyllou 2000) and TOPSIS (Olson 2004). We use TOPSIS at individual customer level to define customer's affinity and buying criteria. Principally, which customer will buy what product can be specified and determined by MCDM technique and how will a customer react throughout a product life cycle in uncertain environment can be specified using rule based probabilistic behaviours.

We rely on actor-based bottom-up simulation for quantitative analyses where each actor responds to the received event based on its own state. The response could either be changing own state or sending a message to some other actor. We evaluate the future behaviour of the system by introducing the notion of 'time' (*i.e.*, simulation 'tick') and understand the consequences of local changes introduced in actor behaviours by observing the overall system behavior over the time.

From methodological perspective, we considered an extended form of a method used extensively in simulation research (Sargent 2005). In our method, we consider a two-step process based on three distinct representations namely problem entity, conceptual model and computerized model. A problem entity, *i.e.*, a real CSP and its environment, is first represented using a high-level conceptual model to formulate the problem statement precisely. We consider this representation as conceptual digital twin of a real system. A conceptual digital twin is then translated into machine processable form, termed as simulatable digital twin.

We adopt *conceptual validation* and *operational validation* as suggested by Robert Sargent (Sargent 2005) to establish the faithfulness of the constructed model. The conceptual model validity ensures that the relevant elements and their relationships are sufficiently identified and described. We involve domain experts for conceptual validity. The operational validity ensures the faithfulness of simulatable digital twin where we rely on historical data, *i.e.*, simulating known historical scenarios and comparing simulation results with historical outcomes through operational graphics and animation of simulation traces. We synchronize simulatable model (*i.e.*, populate simulatable model with the data from actual system) to evaluate adaptation, transformation and design alternatives. The rest of this section illustrates the conceptual digital twin and simulatable digital twin of CSP that lead to quantitative evaluation.

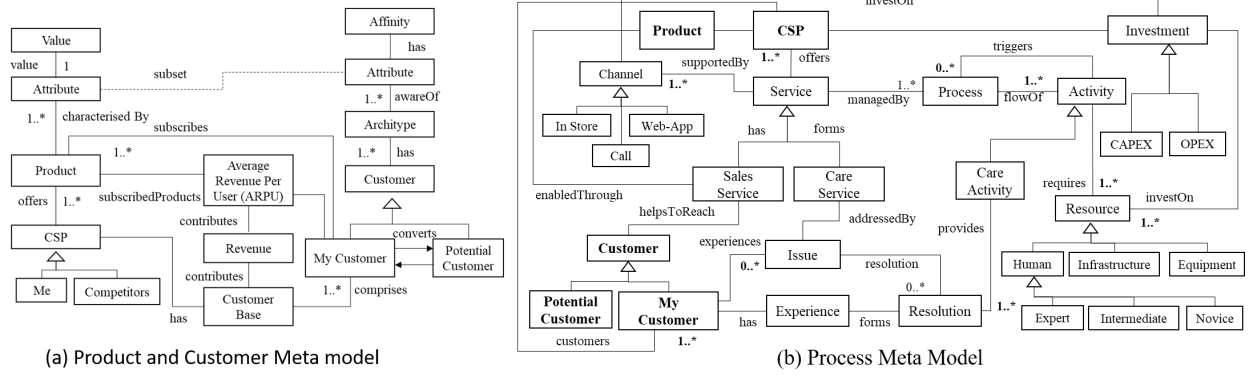


Figure 2: Product and Customer Meta model

4.2 Conceptual Digital Twin

CSP and its environment can be described using four concepts, namely *Product*, *Customer*, *Process* and *Environment*, and their relationships as shown in Figure 1a. These concepts and their relationships are represented using two meta-models with overlapping concepts as shown in Figure 2. As shown in Figure 2a, a CSP offers a set of *Products* with varying characteristics to (existing and potentials) customers. From the perspective of evaluation, we consider two types of CSP: *Me* CSP (i.e., the CSP under consideration) and *Competitor* to represent the competitors of *Me* CSP.

Product is a subscription plan for using a set of CSP offerings for a specific time duration (e.g., month and year) optionally with a device. Therefore, a *Product* can be specified in terms of *Values* of *Attributes* such as international and domestic calls, data, text and movie subscription. The individuals who subscribe one or multiple products offered by a CSP constitute its *Customer Base*. *Customers* may have different preferences or affinity for different product attributes and/or their values thus leading customer *Archetypes*. Typically, the propensity of selecting a specific product by an individual customer depends on the affinity of the individual customer for all product offerings from all existing products of *Me* CSP and its competitors. We use multi-criteria decision-making to compute this propensity. In this formulation, the association of customers from a CSP defines *customer base* KPI, the revenue earned by a CSP through the product subscriptions defines the *revenue* KPI, and revenue per customer helps to compute ARPU of a CSP.

Meta model of Figure 2b depicts *Services* and their relationships with *Customer* and *Product*. A CSP supports two types of services: *Care Service* and *Sales Service*. These services are typically enabled through three types of channels: *Web*, *Call* and *In Store*. Each channel has its own characteristics in terms of availability and reachability to the customers and prospects. *Investment* in terms of *CAPEX* and *OPEX* captures the extent and sophistication of support for each channel.

A CSP targets customers and prospects through *Sales Services*, such as product sell and fulfillment of a subscription. *Care Services* address customer concerns such as billing and network related issues. Both of these services are essentially *Processes* and they can be viewed as orchestration of *Activities* involving a quantum of effort and consuming a quantum of *Resources*. We classify *Resource* into multiple types, such as *Human Resource*, (IT/automation) *Infrastructure* and *Equipment* (e.g., cables and network devices), where *Human Resources* is further classified into multiple types, such as *Expert*, *Intermediate* and *Novice*.

An *Activity*, i.e., *Care Activity*, of *Care Service* helps to resolve *Issues* reported by customers. Multiple factors such as availability of necessary *Equipment*, sophistication of support *Infrastructure*, and quality of *Human Resources* involved determine quality of *Resolution* of an *Issue*. Overall, the efficiency and availability of *Resources* associated with *Sales Services* and support for *Channels* influence the *Customer Base* of a CSP; the characteristics of *Products* of *Me* CSP are associated with the *Issues* experienced by *My Customers*; the characteristics of *My Customers* are linked with the *Issues* raised by *My Customers*, and associated flows of *Care Service* and availability/expertise of the *Resources* associated with *Care Activities* influence *Resolutions* and customer *Experience*. Therefore, appropriate *Investment* strategy in terms of *CAPEX* and *OPEX* for appropriate *Human Resources* and automation *Infrastructures* supported by an effective strategy to replenish required *Equipment* influence process related KPIs such as customer *Experience*, which further effect CSP level KPIs such as customer churn ratio and NPS score.

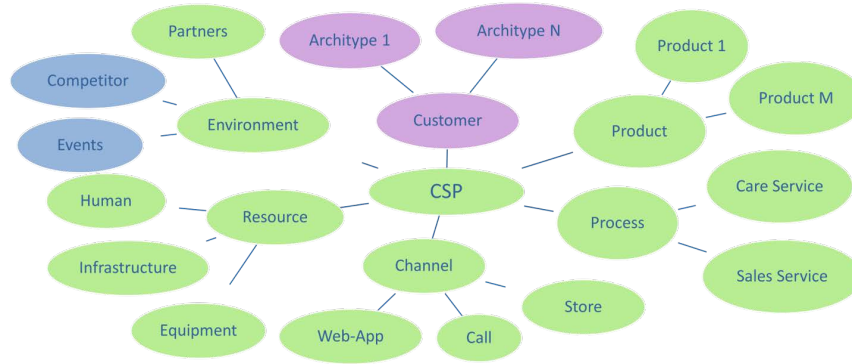


Figure 3: Actor Topology

4.3 Simulatable Digital Twin

We construct simulatable digital twin of a CSP and its environment by transforming all elements defined in conceptual digital twin into actor specification. Primarily, we define a set of actor types (*i.e.*, actor attributes and behavioural specification) and their interactions patterns from above conceptual model. Key actor types for the CSP described above are: *product*, *customer*, *process*, and *resources*. The product actor type is described using a set of tuples of product *Attribute* and *Value* (as discussed in conceptual model). We use event-condition-action (ECA) rules to capture product life-cycle behaviour. The rule such as: for each simulation time ‘tick’ (*i.e.*, event) if the subscription is active (*i.e.*, condition) then reduce remaining subscription time by one time unit. We also define temporal behaviour for billing cycle of a product, and stochastic behaviours for network related issues, billing related issues, and other product characteristics.

Customer actor type defines customer characteristics and two types of behavioural patterns – *subscription behaviour* and *life-cycle behaviour*. Subscription behaviour is the definition of selection criteria of an individual customer, which is defined in terms of personalised affinities for the product attributes. This criterion helps an individual customer to subscribe a product from a list of available options offered by *Me* CSP and its competitors. Life cycle behaviour is the behaviour of an individual during the life cycle of a subscribed product, *i.e.*, how customers react when they face some issues, such as delay in fulfilment of a subscription request, billing related issues, and network outage. We define customer archetypes by specializing these two types of behavioural patterns, *i.e.*, each customer archetype may have different subscription criteria (*e.g.*, budget concern customers look for a product which is more economical, whereas a technology savvy customer chooses a product which offers more data and latest technology/device). Similarly, these customers can react differently when they experience an issue (*e.g.*, not necessary all customers will react same way when they face a network related issue).

In telecom domain, the general consensus about customers is as follows: (a) customers behave almost rationally while subscribing products, however, individual preferences/affinity for product attributes are different and they may exhibit certain level of impulse if selection scores are similar for multiple products (Xiao and Benbasat 2007), and (b) they exhibit some behavioural pattern for the rest of the product life-cycle. For example, while a customer subscribes to a set of products at the same time, however, only some may get renewed and some may get replaced by products offered by competition. We use TOPSIS (Olson 2004) on product attributes and customer characteristics with some randomness to compute which product will be renewed and which substituted by a competitor offering. This helps to mimic rationality infused with a degree of impulse observed in consumer behavior in reality.

We use rule based behaviour with probabilities to specify behavioral patterns of customers and customer archetypes wherein the probabilities are initially learned from the historical data about customer behaviour and then we continuously update those probabilities as simulation progresses to replicate customer behaviour and their adaptation. The processes are represented using composite actor types comprising a set of activity and resource actor types. Essentially, a process is formed by instantiating a process type with appropriate activity actors and resource actors, *i.e.*, human resource actors, actors that represent required automation infrastructure, and actors that represent equipment.

PID	Product Description	PRICE	Voice	Text	Data (GB)	Apple Music	Free Storage (GB)	International Roaming	Additional Data Rate	Customer Archetype	Population (Count)	Price Aware	Storage Seeker	International Traveler	Heavy Data User	Apple Music Enthusiast
1	Start Unlimited	80.00	Unlimited	Unlimited	30	No	No	0	1.5	Impulsive Iggy	770.0	1.0	0.7	0.1	0.9	0.9
2	Play More Unlimited	100.00	Unlimited	Unlimited	100	Rationed	50	0	1	Novice Nigel	783.0	0.9	0.4	0.1	0.6	0.5
3	Do More Unlimited	120.00	Unlimited	Unlimited	120	Rationed	50	0	0.5	Wandering Walter	767.0	0.9	1.0	0.05	0.1	0.1
4	Get More Unlimited	110.00	Unlimited	Unlimited	110	Rationed	75	0	0.5	Loyal Larry	782.0	0.8	0.8	0.05	0.7	0.8
5	Kids Plan	120.00	Unlimited	Unlimited	120	Unlimited	75	0	0.7	Patron Peter	722.0	1.0	1.0	0.1	0.6	0.4
6	International Plan	140.00	Unlimited	Unlimited	90	Unlimited	No	120 (Min./Day)	1.2	Needy Neil	732.0	0.6	0.7	0.1	0.8	0.6
7	Connected Car Plan	150.00	Unlimited	Unlimited	120	Unlimited	100	0	0.8	Discount Dan	758.0	0.0	0.8	0.05	0.2	0.1
(a) Product descriptions										(b) Customer and customer archetype descriptions						
										Unknown 1	738.0	0.9	0.5	0.1	0.3	0.9
										Unknown 2	800.0	0.9	0.6	0.1	0.9	0.9
										Unknown 3	790.0	0.9	1.0	0.1	1.0	0.1

Care Lifecycle	Issue Categories	FTR	Web App	Call	Store	Training	Automation Capex	Automation OPEX	Interaction Impact
Buying	Payment Gateway Related	0.5	5.00	10	20.00	10000	10000	1500	5%
	Price Plan Related	0.7	5.00	10	20.00	20000	25000	2500	10.00%
	Ambiguous TnCs	0.6	8.00	15	25.00	35000	40000	3500	12%
Activation	Activation related	0.8	5.00	10.00	30.00	50000	80000	5000	15%
Usage	Billing Related	0.8	10.00	20.00	30.00	0	0	0	0
	Network Related	0.7	10.00	20.00	40.00	0	0	0	0
	Device Related	0.8	15.00	30.00	40.00	0	0	0	0
	Roaming	0.5	7.00	10.00	40.00	0	0	0	0

(c) Care Related parameters

Figure 4: Product, Customer and Care Configuration

Constructed topology of the actor types is shown in Figure 3. As highlighted using different colours, we use different modelling techniques to represent simulatable digital twin of a CSP. ECA based rule specification (*i.e.*, on receiving of an event if some condition is true for an actor state do a specific action) along with probabilistic distribution for the action space (*i.e.*, do an action with a probability and do other action with other probability) for those actors whose behaviours are known to some extent. Actors that specify products, processes, resources and channels are such type of actors. The behaviours for these actor types are specified by considering the domain understanding and synthesizing available data.

However, we acknowledge that the required understanding for all actor types, especially those which are beyond control of a CSP, might not be available with the CSP. For example, expecting precise understanding to specify the behaviours of all competitors and external events related to network and technology disruptions is not a practical consideration. Here we use neural network models (Specht et al. 1991) trained on market data to represent the behaviour of competitor actors, and historical data centric forecasting models (Ghiassi et al. 2005) for specifying actors that represent external events. Replicating customers of a CSP is the most complex part of simulatable digital twin – we used multi-modelling technique that involve ECA model, historical data centric ML models and MCDM model specifying customers.

We used an actor-based Enterprise Simulation Language (ESL) (Clark et al. 2017) to implement these actor types as ESL: (a) is a scalable actor technology (Barat et al. 2019), (b) supports uncertainty and temporal behaviour in addition to the standard capabilities supported by actor/agent technologies, and (c) is an extensible language that enables seamless integration of MCDM and ML based models.

4.4 Operational Validation and Simulation

Constructed simulatable digital twin of CSP (*i.e.*, virtual environment) is instantiated with real data from the past *i.e.*, inputs and events so as to produce output that matches the past output. Thus, we validate that CSP digital twin is indeed a high-fidelity representation of the real CSP. In this case, we instantiated CSP digital twin as follows: (i) product actors of the *Me* CSP are created by instantiating Attributes and Values of Customer actor type from product catalog (each product is represented as an individual product actor) as shown in Figure 1b, (ii) products of all relevant competitors are also represented as a set product actors by exploring market research data, (iii) customer actors are instantiated from customer records (each customer is represented as an actor with individual characteristics), and (iv) all probabilities of customers and processes actor types are assigned by analyzing the historical traces of customer record and product subscription data. Instantiated simulatable model is validated using three steps operational validity – (a) setting digital twin to a historical state (lets say state of the CSP and its environment as they were before three months from current date), (ii) simulating CSP till date by introducing time events (*i.e.*, ‘Days’), and (iii) comparing simulated states with current state.

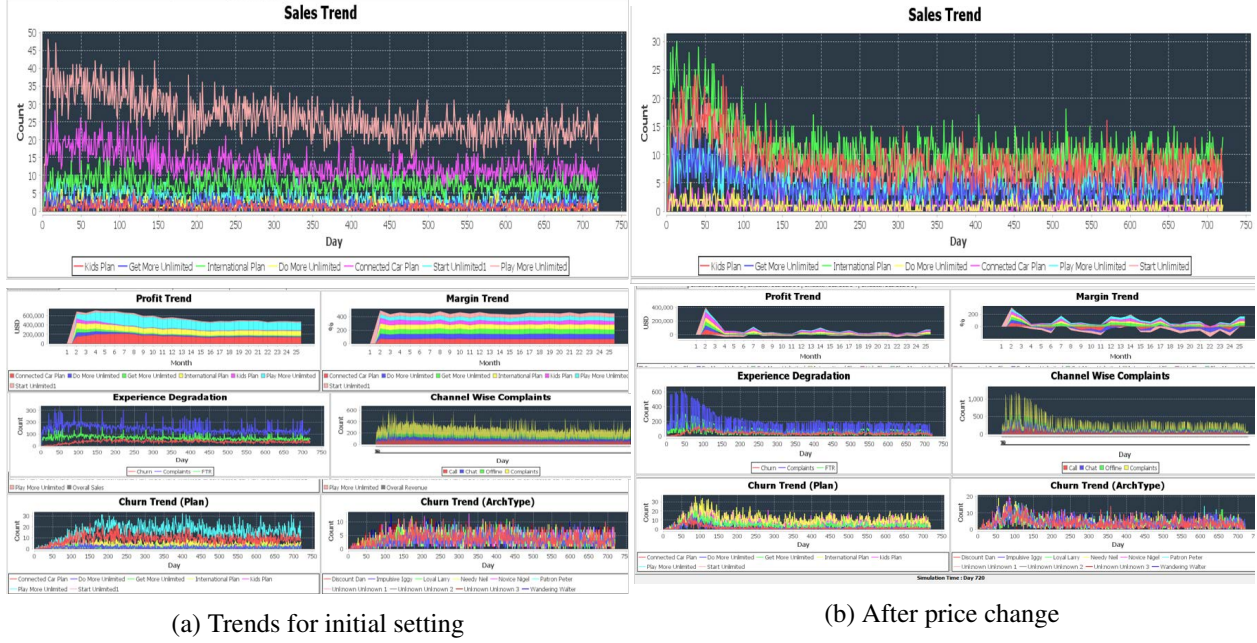


Figure 5: Simulation Dashboard

5 CASE STUDY

We illustrate the utility and efficacy of our simulatable actor-based digital twin approach for informed evidence-driven decision-making of a CSP. In our evaluation, we considered a CSP (*i.e.*, *Me* CSP) whose key focus is to increase customer base, reduce customer churn, improve profit margin and improve customer experience by introducing and altering existing products, and effectively adapting supported sales and care services. From process perspective, we considered sales and care services that include order fulfillment, product activation and usage related issue resolution support through three channels namely: call, web and in store support. We evaluated following types of business scenarios using simulatable digital twin – (i) what would be the impact in subscription pattern when there is a change in the value of one or more product attribute(s) from *Me* and my competitors, (ii) what would be the impact in subscription pattern when a new product is launched, (iii) what would be the near and long term implication (in terms of customer churn rate and customer experience) when there is any change in sales or care services.

In this paper, we illustrate a representative (due to space and confidentiality factors) but close to real business case for exploring product attributes and effectively adapting care process to improve customer base, revenue and customer experience, and reduce customer churn. Consider that the Wimbledon Open (an environmental event) is coming up in UK and all US based CSPs are expecting that many of their customers will travel to UK during the event. But the *Me* CSP sees that its plan (product) that offers international roaming is not popular amongst the offered plans. Now the aim of *Me* CSP is to increase the subscription of its international plan by next two quarters such that it can gain competitive advantage during the up-coming event.

For conducting this illustrative experiment, we constructed *Me* CSP with 7 products, 7642 customers categorized into 10 customer archetypes as shown in Figure 4. We first simulated constructed digital twin and observed the key KPIs to establish the faithfulness of the constructed digital twin. The KPIs, shown in Figure 5a, include the trends of: sales of all 7 products (Figure 5a), revenue margin, customer churn, and customer experience (*i.e.*, how many customers experienced issues, how many complaints are reported, how many of them are resolved first time, *etc.*). As shown in the graphs, the sale of international plan (represented using green colored line in Figure 5a) is not a popular product amongst other products. As shown in Figure 5a, the other KPIs, *i.e.*, customer churn and customer experience (*i.e.*, first time resolution – FTR), are moderate and under control for the current setting – these KPIs and their trends depict the faithfulness of the constructed digital twin with respect to the real CSP. We consider these values as the base for our experimentation.

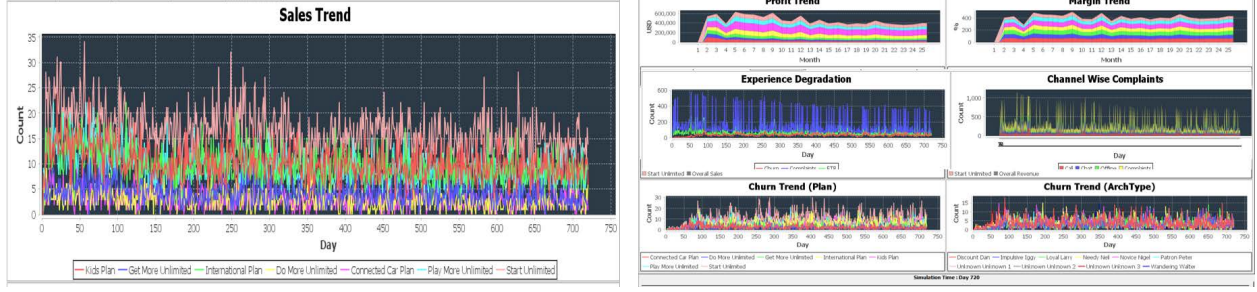


Figure 6: KPIs after improving care services

Now *Me* CSP would like to popularize its international plan with appropriate intervention. There are several interventions possible along price point and CSP related services, such as additional data, cloud storage, subscription to Apple music and various coupons. Here we explore price point as an illustration. We re-simulated constructed digital by changing the monthly subscription rate of international plan from \$140 to \$100 and remaining other attribute values unchanged. The consequence of the intervention is shown in Figure 5b, we can observe that the initial sale of international plan is high for some time right after the intervention is introduced (local optima). However, it is observed that the sale trend of all products, including international plan, is reducing over the time (significantly after 60 ‘Days’, *i.e.*, simulation ‘tick’). Simulation also shows that the profit margin and revenue are reducing and churn rate is increasing as intervention is introduced. Therefore, it indicates that the above intervention is not a robust intervention this CSP. The customer experience graph shows significant drop in FTR, which explains the reason for this bad situation. Essentially, large segment of the existing customers (from price concern archetypes) have started switching from their existing plan to international plan. However, it is observed that the current care process is not capable of supporting such large customer base for international plan (as reflected in FTR). This lack of customer care support for a specific plan results into poor customer experience, which lead to high customer churn. This explains a case where an inappropriate intervention can lead to considerable and irreversible loss for a CSP (a butterfly effect) unless it is backed with appropriate change throughout the CSP, *e.g.*, appropriate change in care and other services.

Next we explore the required appropriate interventions in care service to support increased customer base. As shown in Figure 4c, the *Me* CSP uses robotic automation aids, such as automated voice and video service, to answer certain types of customer calls (*i.e.*, payment gateway, plan price, ambiguous terms and condition, *etc.*) with certain CAPEX and OPEX investments. Now, we explore the consequence if we increase automation support across all types of call by investing appropriate CAPEX and OPEX. The KPIs for one such scenario where investment is made to automate 20% of the customer care related interactions is shown in Figure 6. The sale of international plan become double from initial state (*i.e.*, base scenario) without any degradation in sale over time. The churn ratio is decreased, and profit margin is improved significantly from the earlier experimental result, *i.e.*, experiment with reduced price. However, we can see from the simulation result that the automation related change is not effective for improving the overall profit and revenue despite of improvement in customer base from the base scenario. The key reasons are - price of international plan is less, moreover the CAPEX and OPEX are more to support automation, which are not compensated by revenue improvement due the plan switch. Several experimentations to improve the sale of international plan targeting long term top line and bottom line improvement can be experimented – price point can be explored (*i.e.*, price value between \$100 to \$140), the degree of investment and automation support for different types of interactions can be observed, and popularizing less expensive channels such web chat as opposed to expensive call or in store support can considered as candidate alternatives. We argue that several such in-silico experiments help to understand the short- and long-term consequence of various changes of a CSP, which leads to an evidence driven informed decision-making.

6 CONCLUDING REMARKS

Modern CSPs are complex dynamic systems where they continuously innovate their product offerings, and adapt sales and care services to stay ahead of the competition. In this endeavor, they frequently introduce new products, promotions and offering through appropriate channels, and make course correction on sales and customer care services. The current practice of meeting these business needs is experience-centric and

intuition-driven. This paper presented a novel use of multi-modelling based simulatable digital twin for in-silico quantitative exploration of adaptation and design alternatives of a CSP to help business experts to arrive at the right interventions at right time and with right sequence with precision. A conceptual model to describe products, processes, customer and environment of a CSP is presented. A schema to construct simulatable digital twin is discussed, how simulatable digital twin can help business experts to explore adaptation and design alternatives with precision is demonstrated, and how our approach is applied to business context is briefly mentioned in this paper. We piloted our approach for a geographic segment of a CSP having 46 products, 20 thousand customers classified into 10 different customer archetypes, and 3 competitors with similar range of product offerings. We have started expanding the scope to cover hundreds of products and millions of customers, and early results are quite encouraging. The key observations are: (a) in congruence with the engineering and the field of mission-critical systems, the digital twin based explorations for adaptation and design alternatives seems an effective and economical option for Telecom domain as well, (b) the notion of actor seems to be a pragmatic consideration for representing relevant entities of CSP and its environment, (c) a multi-modelling approach is necessary to comprehensively model CSPs, and (d) establishing faithfulness of the constructed digital twin is a necessary step for which rigorous validation techniques are required. Overall, our approach enables a quantitative, evidence-driven, informed decision-making of large CSP. However, we found that involvement of domain experts is critical to validate conceptual digital twins. Our experimentation revealed that deciding next best exploration option from a large set of alternatives is another area where some improvement is critical so as to reduce analysis and synthesis burden on human experts. As future work, we will be working on two enhancements: (a) a visual domain specific language (DSL) to construct conceptual digital twin – this will help domain experts to specify desired characteristics without much involvement of technical experts, and (b) introduction of learning technique, such as reinforcement learning, to serve as a recommendation system for what action to take next. We also plan to apply our approach to design and adapt products and processes from other consumer facing business domains such as banking and insurance.

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